Federated Latent Factor Learning for Privacy-Preserving Spatio-Temporal Signal Recovery

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Abstract

Wireless sensor network (WSNs) stands out as a burgeoning and promising domain in intelligent sensing. Owing to various factors such as sudden sensor malfunctions or deliberate shutdown of partial nodes to save energy, the collected sensing signals from WSNs commonly have massive missing data, leading to adverse effects on subsequent analysis or decision-making. Latent factor learning (LFL) has proven to be highly effective in recovering the missing data for WSNs. However, the existing LFL models require the collected sensing signals to be maintained in one central place like a central server, which is becoming unacceptable for data owners who are getting increasingly privacy-sensitive. To address this issue, this paper innovatively proposes a **f**ederated **l**atent **f**actor **l**earning (**FLFL**) model for privacy-preserving spatio-temporal signal recovery. Its main idea is two-fold: 1) it designs a sensor-level federated learning framework based on LFL, where each sensor only needs to upload gradient information rather than raw data for training a privacy-preserving recovery model, and 2) it incorporates the spatio-temporal correlation into the designed federated learning framework as the regularization constraint to improve its recovery accuracy. With such designs, FLFL can not only accurately recover the missing data of WSNs but also ensure data owners' privacy-preserving of raw data. To evaluate the proposed FLFL model, extensive experiments have been conducted on four real-world WSNs datasets. The results demonstrate that FLFL significantly outperforms five state-of-the-art federated signal recovery models in terms of recovery accuracy with privacy-preserving.

CCS Concepts

• **Security and privacy → Privacy protections**.

Keywords

Wireless Sensor Networks, Latent Feature Analysis, Spatio-Temporal Correlation, Data Recovery, Federated Learning, Privacy-Preserving

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1 INTRODUCTION

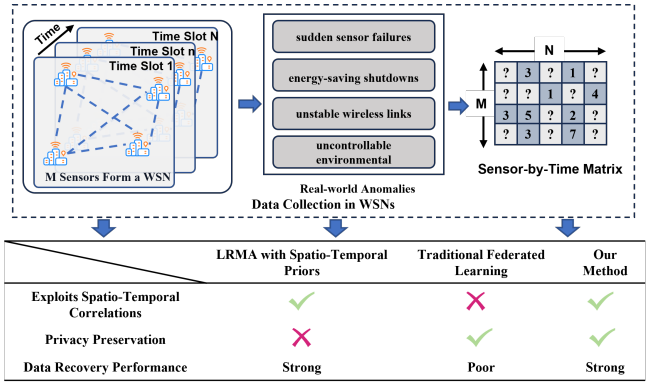

Figure 1: Comparison of Our Method with Spatio-Temporal Prior LRMA and Traditional Federated Learning in WSNs Missing Data Recovery

Wireless sensor networks (WSNs) consist of large numbers of distributed sensor nodes that perform targeted monitoring and sensing tasks [17]. Driven by the growth of big data and advances in sensing, embedded computing, and wireless communication, WSNs are widely deployed in smart cities [10, 23, 36], industrial automation [49], environmental monitoring [16], bridge health monitoring [30], and energy management [28]. Their key advantage is the ability to collect multi-source, heterogeneous environmental and state data at low cost, in near real time, and over long durations [2, 51]. These properties make WSNs a rapidly evolving area of broad interest.

In real deployments, WSNs data often contain missing entries and anomalies. Causes include sudden sensor failures, energy constraints that force nodes offline, unstable wireless links, and uncontrollable environmental factors [7]. Because nodes are typically

fixed in place and sample at preset intervals, observations from neighboring nodes and adjacent time points exhibit strong spatial and temporal correlations. For a single monitoring indicator, the data can be represented as a sensor-by-time matrix whose entries record the observation of a node at a time [33, 34, 40]. Owing to temporal smoothness and sensor redundancy, such data commonly display a low-rank structure together with significant spatio-temporal dependence. As a result, low-rank matrix approximation (LRMA) methods [22, 26] that incorporate spatio-temporal information have become a mainstream approach for imputing missing values, improving data quality, and strengthening the robustness of downstream tasks[38].

Meanwhile, privacy and regulatory requirements are becoming increasingly stringent [11, 47]. Regulations such as the EU General Data Protection Regulation impose clear limits on the collection, processing, and sharing of personal data, making data owners reluctant to exchange raw observations [43]. In city-scale or cross-institution WSNs deployments, data held by different parties are difficult to centralize. This challenges conventional centralized representation-learning pipelines. The core question is: under privacy and compliance constraints and without exposing raw data, how can we still model the intrinsic spatio-temporal correlations of WSNs to achieve accurate missing-data recovery and efficient representation learning?

Federated learning (FL) has gained significant attention as a privacy-preserving paradigm for collaborative modeling [9, 31, 46]. FL trains models locally on distributed data and aggregates model parameters or gradients instead of raw samples, thereby extracting shared knowledge [4, 45]. It has shown promise in recommendation systems[8, 12], intelligent medical diagnosis[19], and wireless communications[5]. However, generic FL frameworks fall short for WSNs specific data recovery. In particular, they provide limited support for strong spatio-temporal correlation modeling, and simple federated strategies struggle to exploit node-level spatio-temporal priors. Therefore, WSNs-oriented federated optimization methods are needed that explicitly encode and leverage spatio-temporal priors. Such methods can enable more robust missing-data recovery and more efficient representation learning under privacy-aware settings.

To address this challenge, this paper innovatively proposes a **f**ederated **l**atent **f**actor **l**earning for privacy-preserving spatio-temporal signal recovery model, named **FLFL**, to recover missing data based on partially observed data mixed with privacy-preserving in WSNs. Its main idea is two-fold: 1) We design a sensor-level federated learning framework based on LFL, where each sensor only needs to upload gradient information rather than raw data for training a privacy-preserving recovery model. 2) FLFL incorporates the spatio-temporal correlation into the designed federated learning framework as the regularization constraint to improve its recovery accuracy. With such designs, it possesses the merits of both high accuracy and privacy-preserving during data recovery. The main contributions of this work are as follows:

- An FLFL model is proposed. It can accurately recover missing data based on partially observed data mixed with privacy-preserving in WSNs.
- To effectively capture spatio-temporal dependencies in FLFL, we propose a greedy region-partitioning strategy for WSNs that groups sensor nodes into multiple spatially proximate and structurally coherent subregions.
- Theoretical analyses and algorithm design are provided for the proposed FLFL model.
- Proofs that gradient information will not be leaked are provided for the proposed FLFL model.

Besides, extensive experiments on four real-world WSNs datasets are performed to evaluate the proposed FLFL model, including comparison with state-of-the-art models and analyses of its characteristics. The results verify that FLFL has significantly higher recovery accuracy than the comparison models. The source code and datasets are available at link: https://github.com/xirimuli/FLFL.git

2 RELATED WORK

2.1 Latent Feature Analysis

The LRMA is widely used for data modeling in WSNs. Because sensors are fixed in space and sample over long horizons at predefined intervals, measurements exhibit inherent temporal and spatial correlations [24, 29, 32, 42]. Adjacent timestamps and nearby sensors often show similar patterns. LRMA therefore represents WSNs data as a matrix: rows correspond to sensor nodes, columns to time steps, and entries to observed readings. To improve recovery accuracy, training typically embeds spatio-temporal priors [41, 48]. Common choices include neighborhood regularization [29] and graph-based penalties that exploit the graph structure of many spatio-temporal signals [6, 20, 27]. For example, Qiu et al.[32] employs a graph Laplacian to promote smoothness on the network, while Mao et al.[24] introduces a hybrid graph regularizer to capture both global and local smoothness.

However, all the above LRMA-based models are centralized. They cannot preserve the privacy and security of data, which is unacceptable for more and more privacy-sensitive sensors. In comparison, the proposed FLFL model is an FL-oriented LRMA approach, which can not only well incorporate the spatio-temporal correlation into LRMA to enhance the accuracy of data recovery, but also well preserve the privacy and security of data in representing WSNs.

2.2 Federated Learning

FL is a distributed training paradigm introduced by Google [11]. It enables multiple participants to jointly train a shared model while keeping raw data local, thereby strengthening privacy protection [25, 44]. In FL, clients update model parameters on-device and share only model updates with a coordinating server, avoiding centralization of personal training data. Privacy concerns have spurred a surge of FL methods. Among them, matrix factorization (MF) within federated frameworks has received particular attention. Chai et al. proposed a secure MF approach under FL, *FedMF* [3]. Lin et al. developed a federated recommender based on MF with a hybrid imputation strategy, *FedRec* [14]. Liang et al. further enhanced *FedRec* by introducing denoising clients to mitigate noise in a privacy-aware manner, *FedRec++* [13]. Lin et al. also integrated meta-learning into MF for federated recommendation, *MetaMF* [15]. Beyond MF, work has explored broader techniques to improve FL utility under privacy constraints; for example, Wu et al. presented

a privacy-preserving federated framework for graph neural network–based recommendation, *FedGNN* [37].

Nevertheless, when applying the aforementioned federated learning model to WSNs, we observed a lack of sufficient consideration for the inherent characteristics of WSNs data, specifically its spatio-temporal correlation. This resulted in suboptimal performance in data recovery. To address this challenge, we propose a novel model called FLFL. In comparison to conventional models, the FLFL model adeptly incorporates spatio-temporal correlation into the model while preserving data privacy. This clever integration significantly enhances the model's accuracy in data recovery.

3 PRELIMINARY

3.1 Problem of Data Recovery in WSNs

The primary challenge in WSNs data recovery lies in reconstructing the missing data using the available observed data [35]. In the following, we provide a formal formulation of the problem.

*Definition 3.1 (**Problem of Data Recovery in WSNs**).* Let $Y \in \mathbb{R}^{M\times N}$ denote the data matrix generated by M sensors over N time slots, where $y_{i,j}$ represents the observed value of sensor i at time j. Denote the sets of observed and unknown entries as Y_K and Y_U, respectively. The recovery task is to estimate Y_U from Y_K. The *sampling rate* is defined as the ratio $|Y_K|/|Y|$.

3.2 LFA-Based Data Recovery in WSNS

The LFA model originates from matrix factorization methods. Given an incomplete data matrix, LFA learns two low-rank latent feature matrices based only on the observed entries [39]. We now present the formulation of LFA for data recovery in WSNs.

Definition 3.2 (LFA-Based Data Recovery in WSNs). The goal of LFA is to estimate the missing set Y_U by factorizing Y into two low-rank latent feature matrices $P \in \mathbb{R}^{M\times k}$ and $Q \in \mathbb{R}^{N\times k}$ with $k \ll \min\{M, N\}$. The reconstructed matrix is approximated as

$$Y \approx PQ^T. \tag{1}$$

The optimization process relies on minimizing the discrepancy between the known entries Y_K and their reconstructed counterparts. Most existing studies employ the squared L_2 loss, while Tikhonov regularization is introduced to alleviate overfitting. The resulting objective function is formulated as

$$\varepsilon(P,Q) = \left\|(Y - PQ^T)\circ\Omega\right\|_2^2 + \lambda\left(\|P\|_F^2 + \|Q\|_F^2\right), \tag{2}$$

where $\Omega \in \{0,1\}^{M\times N}$, with $\Omega_{ij} = 1$ if y_{ij} is observed, and $\Omega_{ij} = 0$ otherwise. The symbol $\circ$ denotes the Hadamard product, $\|\cdot\|_2$ is the matrix L_2-norm, and $\|\cdot\|_F$ represents the Frobenius norm.

3.3 Federated Learning

Definition 3.3 (Federated Learning). Suppose the FL system has M sensors, each of which has a local training dataset S_i, where $i = 1, 2, \ldots, M$. Let S denote the joint training dataset, where $S = \bigcup_{i=1}^{M} S_i$. The M sensors aim to collaboratively train a shared machine learning model based on S.

FL can be generally divided into three categories according to the distribution of data across the feature and sample spaces: horizontal FL, vertical FL, and federated transfer learning [43]. Since all sensors in the WSNs considered in this study generate data with identical feature spaces but differ in sample distributions, the corresponding FL setting falls into the category of horizontal FL. The training process of the FL system usually contains the following four steps:

(1) The server initializes the global model parameters and sends them to each sensor.
(2) Each sensor does not need to share its own data, but it trains a local model based on its data and the global model parameters. After training is complete, the sensor sends the updated gradients to the server.
(3) The server aggregates the updated gradients collected from the sensors to update the global model parameters and then sends the updated global model parameters back to each sensor.
(4) Repeat steps two and three until the model converges or the maximum number of training rounds is reached.

4 METHODOLOGY

4.1 The Framework of FLFL

The FLFL model is a collaborative training–shared LFA model in WSNs. Unlike models based on centralized learning, the FLFL model does not require sensors to exchange raw data, thus preserving the privacy of sensor data. Additionally, considering the spatio-temporal correlation inherent in WSNs data, the FLFL model incorporates this relationship into its training to enhance its ability to recover data, as shown in Figure 2

For each sensor i, the latent feature vector $P_{i,\cdot}$, the Laplacian matrix of the region to which it belongs, and the required local data for training are retained on the device. Sensors within the same region are allowed to exchange their latent feature vectors $P_{j,\cdot}$ $(i \neq j)$, thereby facilitating regional collaboration. In contrast, the server maintains only the global latent feature matrix Q. The overall framework coordinates interactions between the server and sensors to guarantee consistency, efficiency, and security, which is achieved through the following five steps.

(1) In the initialization phase, the WSNs is partitioned into multiple regions using a greedy strategy. The server randomly initializes the global latent feature matrix Q, while each sensor initializes its own latent feature vector.
(2) Each sensor downloads the current Q from the server.
(3) Each sensor performs local training using: its local data, the downloaded Q, its own latent feature vector, the Laplacian matrix of its region, and the latent feature vectors of neighboring sensors within the same region.
(4) After local training, each sensor updates its latent feature vector and uploads the corresponding gradients to the server.
(5) The server aggregates the received gradients and updates the global latent feature matrix Q accordingly.

4.2 Spatio-Temporal Correlation in FLFL

The spatio-temporal correlation of WSNs exhibit two fundamental characteristics: spatial smoothness and temporal smoothness. Spatial smoothness implies that data collected from sensors in close geographical proximity tend to be similar to each other at a given time. Temporal smoothness, on the other hand, signifies that data

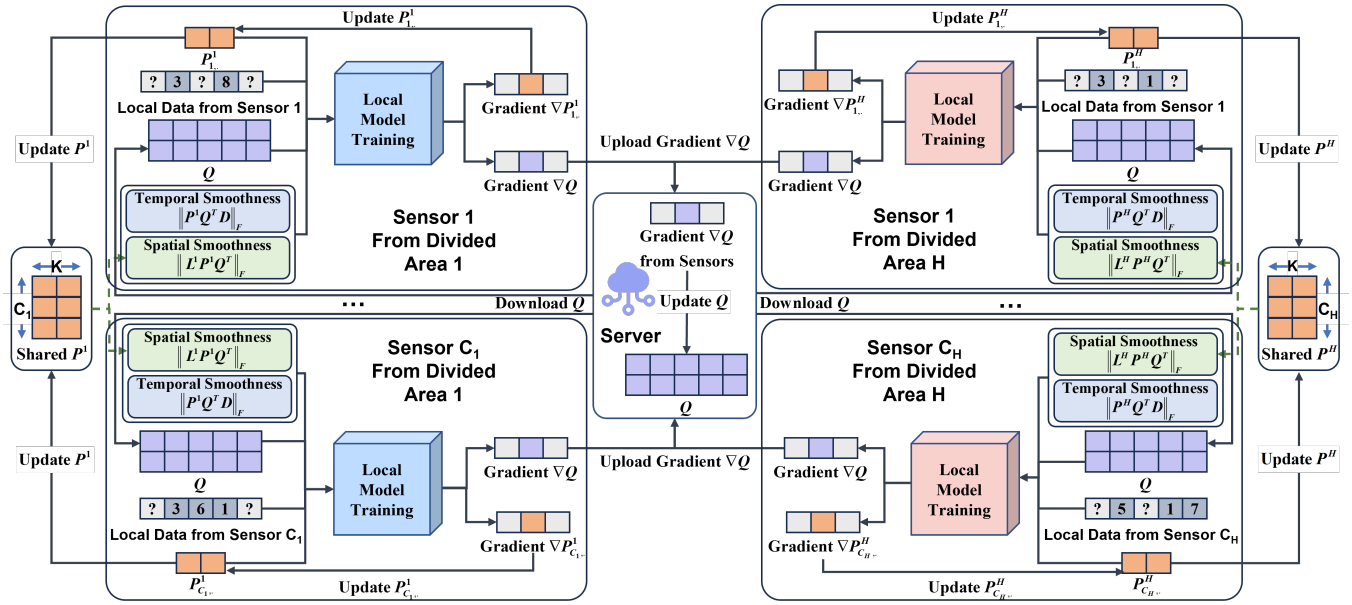

Figure 2: The Framework of FLFL: WSNs are first partitioned into multiple regions using a greedy strategy. Within each region h, the sensors share a regional latent feature matrix P^h. Each sensor then integrates the downloaded latent feature matrix Q from the server with its local data for training, where spatio-temporal smoothness constraints are incorporated. Finally, the gradients with respect to Q are uploaded to the server to update the global latent feature matrix Q.

gathered from the same sensors exhibit smooth variations over time.

4.2.1 Spatial smoothness. Spatial relationships between sensors in WSNs are modeled by an undirected weighted graph $G = (V, E, W)$. V is the set of sensors, with one vertex per sensor. E is the set of undirected edges that encode pairwise associations. $W \in \mathbb{R}^{|V|\times|V|}$ is the weighted adjacency matrix, where $w_{i,i'}$ quantifies the strength of the relationship between vertices i and i'.

Based on sensor coordinates, we compute a distance matrix F. Its entry $f_{i,i'}$ is the Euclidean distance between sensors i and i'. Using F, we build a k-nearest-neighbor graph and define W as

$$w_{i,i'} = \begin{cases} \dfrac{1}{f_{i,i'}^2}, & \text{if } i' \in \mathrm{Top}_k(i), \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

Here, $\mathrm{Top}_k(i)$ is the set of the k nearest neighbors of vertex i.

Given the characteristics of sensor data and privacy requirements, we partition the space into H regions. For each region, we construct an undirected, weighted connectivity graph $G_h = (V_h, E_h, W_h)$, $h = 1, 2, \ldots, H$. Here, V_h is the set of sensors in region h, and $C_h = |V_h|$ denotes its size. The total number of sensors satisfies

$$M = \sum_{h=1}^{H} C_h. \tag{4}$$

Then, We use a greedy strategy to assign the M sensors to H regions. Each region admits at most $Nums_{\max}$ sensors. The goal is to keep sensors within the same region spatially close. The assignment is formulated as

$$\begin{aligned} &\min_{\{V_h\}_{h=1}^{H}} \sum_{h=1}^{H} \sum_{\substack{i,i' \in V_h \\ i<i'}} f_{i,i'} \\ &\text{s.t.} \quad |V_h| \le Nums_{\max} \ (h = 1, \ldots, H), \bigcup_{h=1}^{H} V_h = \{1, \ldots, M\}, \\ &V_h \cap V_{h'} = \varnothing \ \ (\forall\, h \ne h'). \end{aligned} \tag{5}$$

Besides, let L_h denote the Laplacian matrix of graph G_h. Then, L_h is defined as follows: $L_h = \Lambda_h - W_h$, where Λ_h is the diagonal degree matrix of G_h, with entries $D_h(i,i) = \sum_{i'} w^h_{i,i'}$.

$$L^h = \begin{bmatrix} \sum\left(w^h_{1,\cdot}\right) & -w^h_{1,2} & \cdots & -w^h_{1,C_h} \\ -w^h_{2,1} & \sum\left(w^h_{2,\cdot}\right) & \cdots & -w^h_{2,C_h} \\ \cdots & \cdots & \cdots & \cdots \\ -w^h_{C_h,1} & -w^h_{C_h,2} & \cdots & \sum\left(w^h_{C_h,\cdot}\right) \end{bmatrix}_{C_h \times C_h} \tag{6}$$

where $\sum(w^h_{i,\cdot})$, $i \in \{1, 2, \ldots, C_h\}$, denotes the sum of the i-th row of the matrix W_h. Then, the spatial smoothness of the reconstructed matrix G_h is guaranteed by minimizing the following regularization constraint: $\left\|L_h P^h Q^T\right\|_F^2$.

4.2.2 Temporal smoothness. Since each row of the WSNs data matrix Y corresponds to the sequential readings of a sensor at fixed intervals, the entries along a row generally follow a smooth temporal pattern. That is, consecutive elements in the same row tend to exhibit only minor variations. Consequently, the differences between adjacent time slots should remain small, and this constraint

should also hold for the reconstructed matrix. To enforce such smoothness, we introduce the temporal difference matrix D defined as follows:

$$D = \begin{bmatrix} -1 & 0 & 0 & \dots & 0 \\ 1 & -1 & 0 & \dots & 0 \\ 0 & 1 & -1 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 1 & -1 \\ 0 & 0 & \dots & 0 & 1 \end{bmatrix}_{N \times N-1}, \tag{7}$$

$\left\|PQ^T D\right\|_F^2$ captures the temporal variation of the reconstructed data. Specifically, it encodes the successive differences between adjacent time slots, expressed as $[(\hat{y}_{i,1} - \hat{y}_{i,0})^2,\ (\hat{y}_{i,2} - \hat{y}_{i,1})^2,\ \dots,\ (\hat{y}_{i,N} - \hat{y}_{i,N-1})^2]$, which quantify the changes of sensor i across consecutive intervals.

4.3 Federated Optimization

In the federated learning framework, each sensor conducts training by leveraging four sources of information: its own local data, the latent feature vectors of other sensors in the same region, the Laplacian matrix of its region, and the global latent feature matrix Q provided by the server. Therefore, the objective function for each sensor i in the h-th region can be constructed as follows:

$$\begin{aligned} \varepsilon\left(p_{i,\cdot}^h, Q\right) = & \sum_{y_{i,j} \in Y_K^i} \left(y_{i,j} - p_{i,\cdot}^h q_{j,\cdot}\right)^2 + \lambda \sum_{y_{i,j} \in Y_K^i} \left((p_{i,\cdot}^h)^2 + (q_{j,\cdot})^2\right) \\ & + z_1 \sum_{y_{i,j} \in Y_K^i} \left(\left(L^h P^h Q^{\mathrm{T}}\right)_{(i,j)}\right)^2 \\ & + z_2 \sum_{y_{i,j} \in Y_K^i} \left(\left(P^h Q^{\mathrm{T}} D\right)_{(i,j)}\right)^2. \end{aligned} \tag{8}$$

where Y_i^K is the known entry set of Y in the i-th sensor, and P_h consists of all sensor latent feature vectors in region h. Notably, the Laplacian matrix L_h in different regions is different. Then, the instant loss $\varepsilon_{i,j}^h$ of $\varepsilon(P, Q)$ for the i-th sensor at the j-th time slot in the h-th region on a single entry $y_{i,j} \in Y_i^K$ can be expressed as:

$$\begin{aligned} \varepsilon_{i,j}^h = & \left(y_{i,j} - p_{i,\cdot}^h q_{j,\cdot}\right)^2 + \lambda\left((p_{i,\cdot}^h)^2 + (q_{j,\cdot})^2\right) \\ & + z_1\left(\left(L^h P^h Q^{\mathrm{T}}\right)_{(i,j)}\right)^2 + z_2\left(\left(P^h Q^{\mathrm{T}} D\right)_{(i,j)}\right)^2. \end{aligned} \tag{9}$$

Our goal is to obtain the latent feature matrix Q stored on the server and the latent feature matrix P stored on the sensors. To achieve this goal, we employ the stochastic gradient descent (SGD) method to solve this optimization problem.

4.3.1 Federated learning in the sensor. In the t-th round of iteration, each sensor i downloads the latent feature matrix Q from the server as well as the local latent feature vectors of other sensors within the same region. Subsequently, sensor i from region h can compute the gradient $\nabla(p_{i,\cdot}^h)^t$ of the local vector and the gradient ∇q^t of Q. According to Eq.(9), $\nabla(p_{i,\cdot}^h)^t$ and $\nabla q_{j,\cdot}^t$, which represents the j-th latent feature vector gradient of ∇q^t, can be computed at the element-wise level as follows:

$$\begin{aligned} \nabla(p_{i,k'}^h)^t = & -2q_{j,k'}^t\left(y_{i,j} - (p_{i,\cdot}^h)^t q_{j,\cdot}^t\right) + 2\lambda(p_{i,k'}^h)^t \\ & + 2z_1\left(\left(L^h(L^h)^{\mathrm{T}}\right)_{i,\cdot}(P^h)^t\,(q_{j,\cdot}^t)^{\mathrm{T}}\right) q_{j,k'}^t \\ & + 2z_2\left((p_{i,\cdot}^h)^t (Q^t)^{\mathrm{T}}(DD^{\mathrm{T}})_{\cdot,j}\right) q_{j,k'}^t, \end{aligned} \tag{10}$$

$$\begin{aligned} \nabla q_{j,k'}^t = & -2(p_{i,k'}^h)^t\left(y_{i,j} - (p_{i,\cdot}^h)^t q_{j,\cdot}^t\right) + 2\lambda q_{j,k'}^t \\ & + 2z_1\left(\left(L^h(L^h)^{\mathrm{T}}\right)_{i,\cdot}(P^h)^t\,(q_{j,\cdot}^t)^{\mathrm{T}}\right)(p_{i,k'}^h)^t \\ & + 2z_2\left((p_{i,\cdot}^h)^t (Q^t)^{\mathrm{T}}(DD^{\mathrm{T}})_{\cdot,j}\right)(p_{i,k'}^h)^t. \end{aligned} \tag{11}$$

where k' is the k'-th element in the latent feature dimension k.

For sensor i from region h, it retains its own latent feature vector $p_{i,\cdot}^h$. After computing the gradients $\nabla(p_{i,\cdot}^h)^t$ and ∇q^t, the latent vector $p_{i,\cdot}^h$ for sensor i at the element-wise level can be updated as follows:

$$\left(p_{i,k'}^h\right)^{t+1} = \left(p_{i,k'}^h\right)^t - \eta\nabla\left(p_{i,k'}^h\right)^t, \tag{12}$$

where η represents the learning rate of SGD, governing the pace of gradient descent iterations. Then, sensor i sends the gradient ∇q^t to the server for updating the latent feature matrix Q. The learning procedure of each sensor is detailed in the Appendix on Algorithm Design.

4.3.2 Federated learning in the server. Once the server receives the gradient ∇q^t at the t-th iteration, it immediately updates the latent feature matrix Q as follows:

$$Q^{t+1} = Q^t - \eta\nabla q^t, \tag{13}$$

So, at element-wise update of Q can be expressed as:

$$(q_{j,k'})^{t+1} = (q_{j,k'})^t - \eta\nabla(q_{j,k'})^t, \tag{14}$$

where $(q_{j,k'})^{t+1}$ is the k'-th element at the j-th time slot in the $(t+1)$-th round of iteration. The learning process of the server is detailed in the Appendix on Algorithm Design.

4.4 Privacy Analysis

Figure 2 illustrates a framework where sensors can utilize their local data to train models and iteratively upload gradient information to the server in plain text format, enabling the updating of the global model. Next, we will demonstrate that a collaborative training–shared LFA model in WSNs can prevent gradient leakage according to [3], thereby safeguarding sensor privacy data and protecting it from intrusion by the server.

For sensor i in region h, let $(p_{i,\cdot}^h)^t$ denote its latent feature vector at iteration t. Let U_i be the set of time slot observed by sensor i. The gradient with respect to the j-th row of Q (for $j \in U_i$) that sensor i sends to the server is given below.

$$\begin{aligned} G_j^t = & -2(p_{i,\cdot}^h)^t\left(y_{i,j} - (p_{i,\cdot}^h)^t q_{j,\cdot}^t\right) + 2\lambda q_{j,\cdot}^t \\ & + 2z_1\left(\left(L^h(L^h)^{\mathrm{T}}\right)_{i,\cdot}(P^h)^t\,(q_{j,\cdot}^t)^{\mathrm{T}}\right)(p_{i,\cdot}^h)^t \\ & + 2z_2\left((p_{i,\cdot}^h)^t (Q^t)^{\mathrm{T}}(DD^{\mathrm{T}})_{\cdot,j}\right)(p_{i,\cdot}^h)^t. \end{aligned} \tag{15}$$

Let the three auxiliary terms $\hat{y}^t_{i,j}$, F^t_1 and F^t_2 be defined as

$$\begin{aligned} \hat{y}^t_{i,j} &= (p^h_{i,\cdot})^t q^t_{j,\cdot}, \\ F^t_1 &= z_1\Big(\big(L^h (L^h)^{\mathrm{T}}\big)_{i,\cdot} (P^h)^t \, (q^t_{j,\cdot})^{\mathrm{T}} \Big), \\ F^t_2 &= z_2\Big((p^h_{i,\cdot})^t \, (Q^t)^{\mathrm{T}} \, (DD^{\mathrm{T}})_{\cdot,j} \Big). \end{aligned} \tag{16}$$

Similarly, in the $(t+1)$-th round of iteration, we can get the following equation:

$$\begin{aligned} G^{t+1}_j = &- 2(p^h_{i,\cdot})^{t+1} \Big[\big(y_{i,j} - (\hat{y}^{t+1}_{i,j}\big) - F^{t+1}_1 - F^{t+1}_2 \Big] \\ &+ 2\lambda q^{t+1}_{j,\cdot}. \end{aligned} \tag{17}$$

According to Eq.(12), the correlation between $(p^h_{i,\cdot})^t$ and $(p^h_{i,\cdot})^{t+1}$ is:

$$\begin{aligned} &(p^h_{i,\cdot})^t - (p^h_{i,\cdot})^{t+1} \\ &= -2\eta \sum_{j \in U_i} \Big\{ q^t_{j,\cdot} \Big[\big(y_{i,j} - \hat{y}^t_{i,j}\big) - F^t_1 - F^t_2 \Big] - \lambda (p^h_{i,\cdot})^t \Big\}. \end{aligned} \tag{18}$$

A closer look at element-wise calculation of Eq.(15)-(18):

$$\begin{cases} G^t_{j1} = -2(p^h_{i1})^t \Big[\big(y_{i,j} - \hat{y}^t_{i,j}\big) - F^t_1 - F^t_2 \Big] + 2\lambda q^t_{j1}, \\ \quad \vdots \\ G^t_{jk} = -2(p^h_{ik})^t \Big[\big(y_{i,j} - \hat{y}^t_{i,j}\big) - F^t_1 - F^t_2 \Big] + 2\lambda q^t_{jk}. \end{cases} \tag{19}$$

$$\begin{cases} G^{t+1}_{j1} = -2(p^h_{i1})^{t+1} \Big[\big(y_{i,j} - \hat{y}^{t+1}_{i,j}\big) - F^{t+1}_1 - F^{t+1}_2 \Big] + 2\lambda q^{t+1}_{j1}, \\ \quad \vdots \\ G^{t+1}_{jk} = -2(p^h_{ik})^{t+1} \Big[\big(y_{i,j} - \hat{y}^{t+1}_{i,j}\big) - F^{t+1}_1 - F^{t+1}_2 \Big] + 2\lambda q^{t+1}_{jk}. \end{cases} \tag{20}$$

$$\begin{cases} (p^h_{i1})^t - (p^h_{i1})^{t+1} = \\ -2\eta \sum_{j \in U_i} \Big\{ q^t_{j1} \Big[\big(y_{i,j} - \hat{y}^t_{i,j}\big) - F^t_1 - F^t_2 \Big] - \lambda (p^h_{i1})^t \Big\}, \\ \quad \vdots \\ (p^h_{ik})^t - (p^h_{ik})^{t+1} = \\ -2\eta \sum_{j \in U_i} \Big\{ q^t_{jk} \Big[\big(y_{i,j} - \hat{y}^t_{i,j}\big) - F^t_1 - F^t_2 \Big] - \lambda (p^h_{ik})^t \Big\}. \end{cases} \tag{21}$$

Now, let us turn to analyze the a-th entry of $p^h_{i,\cdot}$, denoted as $p^h_{i,a}$. According to Eq.(18), we obtain the following equations:

$$\frac{(p^h_{ia})^t}{(p^h_{i,(a+1)})^t} = \frac{G^t_{ja} - 2\lambda q^t_{ja}}{G^t_{j,(a+1)} - 2\lambda q^t_{j,(a+1)}}. \tag{22}$$

$$(y_{i,j} - \hat{y}^t_{i,j}) - F^t_1 - F^t_2 = \frac{G^t_{ja} - 2\lambda q^t_{ja}}{-2\Big(p^h_{ia}\Big)^t}, \tag{23}$$

$$\Big(p^h_{ia}\Big)^t - \Big(p^h_{ia}\Big)^{t+1} = -2\eta \sum_{j \in U_i} \left\{ q^t_{ja} \frac{G^t_{ja} - 2\lambda q^t_{ja}}{-2\Big(p^h_{ia}\Big)^t} - \lambda \Big(p^h_{ia}\Big)^t \right\}, \tag{24}$$

Thus, $(p^h_{i,\cdot})^{t+1}$ can be represented as $(p^h_{i,\cdot})^t$. Therefore, based on Eqs. (19) and (20), we define, for simplicity, O^t as

$$O^t = (p^h_{ia})^t + 2\eta \sum_{j \in U_i} \big(q^t_{ja} \frac{G^t_{ja} - 2\lambda q^t_{ja}}{-2(p^h_{ia})^t} - \lambda (p^h_{ia})^t \big). \tag{25}$$

Then we obtain the following equation:

$$\begin{aligned} &\frac{G^t_{ja} - 2\lambda q^t_{ja}}{-2(p^h_{ia})^t} + F^t_1 + F^t_2 + (p^h_{i,\cdot})^t q^t_{j,\cdot} \\ &= \frac{G^{t+1}_{ja} - 2\lambda q^{t+1}_{ja}}{-2O^t} + F^{t+1}_1 + F^{t+1}_2 + \sum_{a=1}^{k} \Big(O^t \; q^{t+1}_{ja} \Big). \end{aligned} \tag{26}$$

Equation (26) shows that, from the server's perspective, the observed gradients G (together with Q) are entangled with three unknowns that remain local to the region or sensor: (i) the regional Laplacian L_h, (ii) the private latent vector $p^h_{i,\cdot}$ of sensor i, and (iii) the regional latent matrix P^h. Owing to this coupling, inverting gradients to recover the underlying sensor values is ill-posed, thereby strengthening protection against gradient-based privacy leakage.

5 EXPERIMENTS

5.1 Experimental Setup

Table 1: The Properties of Experimental Datasets

No.	Name	Sensors	Time slots	Time	Min./Max.
D1	Beijing CO Concentration	35	8647	2014-05 2015-04	0.1/20.0
D2	Sea Surface Temperature	70	1733	1870-01 2014-12	0.01/30.31
D3	Beijing PM2.5 Concentration	35	8647	2014-05 2015-04	3.0/773.7
D4	Chongqing SO_2 Concentration	36	8670	2022-01 2022-12	1.0/400.0

Datasets.

We evaluate on four real-world WSNs datasets spanning CO, sea surface temperature, PM2.5, and SO_2. A summary of their characteristics is provided in Table 1.

Evaluation Metrics.

We evaluate reconstruction accuracy using the mean absolute error (MAE) and the root mean squared error (RMSE). RMSE squares residuals and therefore emphasizes large deviations, whereas MAE weights all errors equally. Because WSNs measurements inevitably contain outliers, reporting both metrics is informative: RMSE highlights rare large reconstruction failures, while MAE reflects typical entry-wise accuracy. MAE and RMSE are calculated as follows:

$$MAE = \left(\sum_{y_{i,j} \in \gamma} \big| y_{i,j} - \hat{y}_{i,j} \big| \right) \Big/ |\gamma|, RMSE = \sqrt{\left(\sum_{y_{i,j} \in \gamma} \big(y_{i,j} - \hat{y}_{i,j} \big)^2 \right) \Big/ \gamma}$$

where $\hat{y}_{i,j}$ denotes the estimate of $y_{i,j}$ and γ denotes the testing set; lower values of RMSE/MAE indicate better accuracy.

Table 2: The comparison results of recovery accuracy with different sampling rates.

Dataset	Sampling Rate	Metric	RFRec	FeSoG	FedMF	FedRec++	MetaMF	FLFL
D1	0.1	MAE	$0.765_{\pm 0.026}$	$0.434_{\pm 0.004}$	$0.472_{\pm 0.002}$	$0.434_{\pm 0.005}$	$0.381_{\pm 0.001}$	$\mathbf{0.305_{\pm 0.002}}$
		RMSE	$1.808_{\pm 0.044}$	$0.799_{\pm 0.003}$	$0.846_{\pm 0.002}$	$0.782_{\pm 0.003}$	$0.716_{\pm 0.002}$	$\mathbf{0.434_{\pm 0.002}}$
	0.5	MAE	$0.309_{\pm 0.012}$	$0.292_{\pm 0.002}$	$0.305_{\pm 0.002}$	$0.287_{\pm 0.002}$	$0.318_{\pm 0.001}$	$\mathbf{0.201_{\pm 0.000}}$
		RMSE	$0.586_{\pm 0.017}$	$0.578_{\pm 0.002}$	$0.568_{\pm 0.002}$	$0.557_{\pm 0.002}$	$0.622_{\pm 0.001}$	$\mathbf{0.399_{\pm 0.001}}$
	0.9	MAE	$0.284_{\pm 0.021}$	$0.275_{\pm 0.002}$	$0.261_{\pm 0.001}$	$0.250_{\pm 0.000}$	$0.284_{\pm 0.001}$	$\mathbf{0.155_{\pm 0.001}}$
		RMSE	$0.563_{\pm 0.033}$	$0.568_{\pm 0.002}$	$0.529_{\pm 0.002}$	$0.512_{\pm 0.000}$	$0.565_{\pm 0.001}$	$\mathbf{0.328_{\pm 0.001}}$
D2	0.1	MAE	$0.535_{\pm 0.024}$	$1.477_{\pm 0.010}$	$0.922_{\pm 0.018}$	$0.956_{\pm 0.016}$	$0.438_{\pm 0.008}$	$\mathbf{0.324_{\pm 0.002}}$
		RMSE	$0.889_{\pm 0.026}$	$2.132_{\pm 0.016}$	$1.588_{\pm 0.020}$	$1.197_{\pm 0.019}$	$0.591_{\pm 0.010}$	$\mathbf{0.433_{\pm 0.011}}$
	0.5	MAE	$0.172_{\pm 0.019}$	$0.622_{\pm 0.006}$	$0.187_{\pm 0.002}$	$0.198_{\pm 0.001}$	$0.264_{\pm 0.004}$	$\mathbf{0.157_{\pm 0.001}}$
		RMSE	$0.239_{\pm 0.022}$	$0.852_{\pm 0.007}$	$0.253_{\pm 0.001}$	$0.260_{\pm 0.002}$	$0.341_{\pm 0.006}$	$\mathbf{0.215_{\pm 0.001}}$
	0.9	MAE	$0.133_{\pm 0.014}$	$0.565_{\pm 0.003}$	$0.129_{\pm 0.001}$	$0.126_{\pm 0.001}$	$0.192_{\pm 0.004}$	$\mathbf{0.120_{\pm 0.001}}$
		RMSE	$0.179_{\pm 0.018}$	$0.738_{\pm 0.003}$	$0.173_{\pm 0.001}$	$0.169_{\pm 0.001}$	$0.250_{\pm 0.004}$	$\mathbf{0.161_{\pm 0.001}}$
D3	0.1	MAE	$36.272_{\pm 0.621}$	$26.256_{\pm 0.653}$	$27.024_{\pm 0.043}$	$23.697_{\pm 0.053}$	$25.607_{\pm 0.053}$	$\mathbf{18.023_{\pm 0.290}}$
		RMSE	$70.306_{\pm 0.642}$	$46.603_{\pm 0.692}$	$48.078_{\pm 0.033}$	$41.243_{\pm 0.028}$	$43.126_{\pm 0.064}$	$\mathbf{30.479_{\pm 0.128}}$
	0.5	MAE	$17.663_{\pm 1.057}$	$19.461_{\pm 0.752}$	$17.791_{\pm 0.033}$	$16.859_{\pm 0.018}$	$18.631_{\pm 0.069}$	$\mathbf{10.280_{\pm 0.006}}$
		RMSE	$29.977_{\pm 1.076}$	$34.427_{\pm 0.649}$	$30.807_{\pm 0.032}$	$29.896_{\pm 0.016}$	$32.557_{\pm 0.099}$	$\mathbf{19.335_{\pm 0.023}}$
	0.9	MAE	$15.630_{\pm 1.468}$	$18.722_{\pm 0.559}$	$14.486_{\pm 0.012}$	$14.831_{\pm 0.011}$	$16.548_{\pm 0.041}$	$\mathbf{7.993_{\pm 0.001}}$
		RMSE	$26.425_{\pm 0.938}$	$32.675_{\pm 0.571}$	$25.861_{\pm 0.015}$	$26.014_{\pm 0.012}$	$28.389_{\pm 0.114}$	$\mathbf{14.741_{\pm 0.001}}$
D4	0.1	MAE	$7.320_{\pm 0.012}$	$5.268_{\pm 0.830}$	$3.921_{\pm 0.023}$	$3.753_{\pm 0.027}$	$3.274_{\pm 0.014}$	$\mathbf{2.307_{\pm 0.029}}$
		RMSE	$18.102_{\pm 0.017}$	$7.956_{\pm 0.712}$	$6.713_{\pm 0.019}$	$6.526_{\pm 0.022}$	$5.919_{\pm 0.004}$	$\mathbf{4.806_{\pm 0.024}}$
	0.5	MAE	$3.511_{\pm 0.005}$	$5.130_{\pm 0.058}$	$3.548_{\pm 0.009}$	$3.330_{\pm 0.007}$	$3.070_{\pm 0.007}$	$\mathbf{1.830_{\pm 0.002}}$
		RMSE	$8.511_{\pm 0.007}$	$7.955_{\pm 0.054}$	$6.146_{\pm 0.010}$	$5.821_{\pm 0.008}$	$5.701_{\pm 0.006}$	$\mathbf{4.114_{\pm 0.003}}$
	0.9	MAE	$3.692_{\pm 0.008}$	$4.837_{\pm 0.042}$	$3.523_{\pm 0.009}$	$3.382_{\pm 0.006}$	$3.122_{\pm 0.007}$	$\mathbf{1.780_{\pm 0.006}}$
		RMSE	$6.023_{\pm 0.016}$	$7.787_{\pm 0.046}$	$6.123_{\pm 0.008}$	$5.973_{\pm 0.005}$	$5.756_{\pm 0.008}$	$\mathbf{4.000_{\pm 0.008}}$
Statistical Analysis		**Win/Loss**	**24/0**	**24/0**	**24/0**	**24/0**	**24/0**	**120/0**
		p-value	**0.0001**	**0.0001**	**0.0001**	**0.0001**	**0.0001**	**–**
		F-rank	**4.376**	**5.229**	**3.883**	**2.813**	**3.688**	**1**

Baselines.

We compare the proposed FLFL model against state-of-the-art federated learning methods: RFRec[18],FeSoG [21], FedMF [3], FedRec++ [13], MetaMF [15]). See the Appendix for details.

5.2 Performance Comparison

We vary the sampling rate from 0.1 to 0.9 to compare all models. For brevity, Table 2 reports results at sampling rates 0.1, 0.5, and 0.9, while full results for all rates are provided in the Appendix. To assess statistical significance, we conduct win/loss analysis, the Wilcoxon signed-ranks test, and the Friedman test. As shown in Table 2, both MAE and RMSE generally decrease as the sampling rate increases. Two additional findings are noteworthy:

Relative performance. Compared to federated baselines, FLFL achieves the lowest MAE / RMSE in all settings evaluated.

Statistical significance. Across the four datasets, all p-values are below 0.05, and FLFL attains the best (lowest) Friedman rank, indicating statistically significant improvements over federated models.

Overall, FLFL sustains higher accuracy than non-federated spatio-temporal methods and delivers the best recovery performance among federated approaches, attributable to its explicit modeling of spatio-temporal correlations in WSNs.

5.3 Ablation Study

Under a fixed sampling rate of 0.1, we perform a systematic ablation of spatio-temporal correlations in FLFL to assess their contribution to recovery accuracy. Figure 3 reports results for four settings: no spatio-temporal correlation, spatial-only, temporal-only, and joint spatio-temporal modeling. The experiments show that introducing either spatial or temporal correlation alone markedly improves

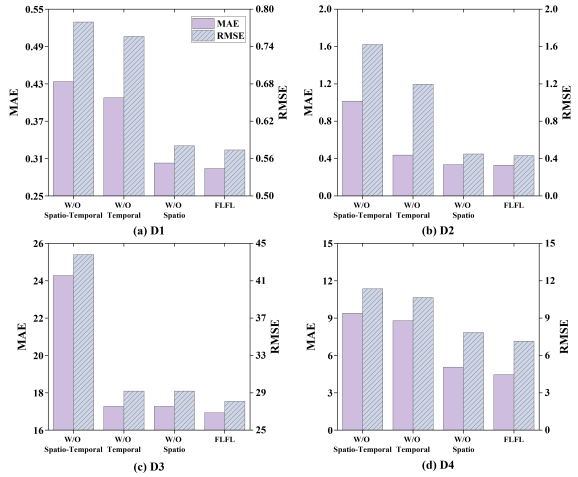

Figure 3: The Ablation study of Spatio-Temporal Correlations

recovery accuracy over the baseline without any correlations; and jointly modeling spatio-temporal correlations achieves the best performance across all datasets, significantly surpassing the single-correlation variants. In sum, explicitly modeling spatio-temporal correlations in FLFL is both necessary and effective for improving missing-data recovery.

5.4 Sensitivity to the Number of Partitioned Regions H

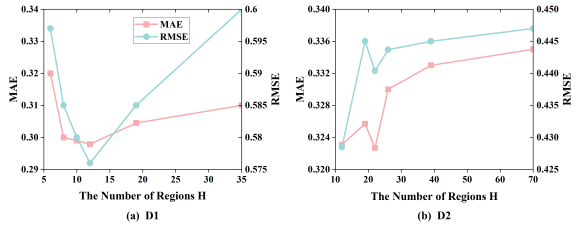

Figure 4: The effect of H on FLFL's recovery accuracy

We fix the sampling rate at 0.1 and hold all other hyperparameters constant. To evaluate the effect of H on FLFL's recovery accuracy, we vary H on datasets D1 and D2; the maximal setting $H = M$ assigns each sensor to its own region. As shown in Figure 4, MAE/RMSE initially decrease with increasing H, attain an optimum, and then rise. This pattern reflects two opposing effects: when H is small, regions are large and each sensor is influenced by many neighbors, effectively injecting additional noise; when H is too large, regions become sparse with fewer neighbors, weakening spatial correlations and degrading recovery. Overall, choosing an appropriate number of partitioned regions is critical for FLFL to fully exploit spatial dependencies.

5.5 Gradient Inversion Attack Experiment

Private information can be inferred via gradient inversion attacks such as Deep Leakage from Gradients (DLG)[50]. To assess the privacy protection of FLFL, we design a DLG-inspired gradient inversion attack that attempts to reconstruct client data from the gradients transmitted to the server. Using the gradients from the final training iteration, we iteratively perform gradient inversion until convergence and obtain reconstructed dummy values. As shown in Figure 5 the reconstructed values exhibit substantial distributional deviations from the ground-truth data across all four datasets, indicating that FLFL effectively protects data privacy and prevents attackers from recovering the original data distribution.

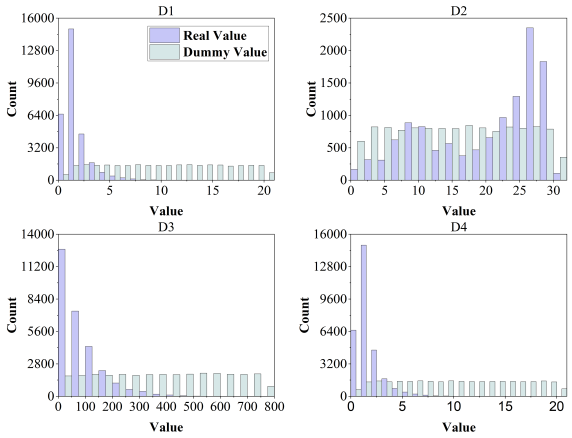

Figure 5: The distributions between real values and dummy values on all datasets.

6 CONCLUSION

This paper proposes FLFL, for privacy-preserving spatio-temporal signal recovery in WSNs. The core idea is twofold: (i) a sensor-level federated learning framework based on LFL, in which each sensor uploads gradient information rather than raw data to train a privacy-preserving recovery model; and (ii) incorporation of spatio-temporal correlation into the federated framework as a regularization constraint to enhance recovery accuracy. This design endows FLFL with both strong privacy protection and high accuracy for WSNs data recovery. Experiments on four real-world WSNs datasets demonstrate that FLFL significantly outperforms related state-of-the-art models in terms of both recovery accuracy and privacy preservation. In future work, we will further optimize FLFL to address outliers arising in WSNs data collection and to improve the model's robustness.

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References

[1] Martin Abadi, Andy Chu, Ian Goodfellow, H Brendan McMahan, Ilya Mironov, Kunal Talwar, and Li Zhang. 2016. Deep learning with differential privacy. In *Proceedings of the 2016 ACM SIGSAC conference on computer and communications security*. 308–318.

[2] Elham Alotaibi, R Bin Sulaiman, and Mohammed Almaiah. 2025. Assessment of cybersecurity threats and defense mechanisms in wireless sensor networks. *J. Cyber Secur. Risk Audit* 2025, 1 (2025), 47–59.

[3] Di Chai, Leye Wang, Kai Chen, and Qiang Yang. 2020. Secure federated matrix factorization. *IEEE Intelligent Systems* 36 (2020), 11–20.

[4] Ruonan Chen, Ye Dong, Yizhong Liu, Tingyu Fan, Dawei Li, Zhenyu Guan, Jianwei Liu, and Jianying Zhou. 2025. FLock: Robust and Privacy-Preserving Federated Learning based on Practical Blockchain State Channels. In *Proceedings of the ACM on Web Conference 2025*. Association for Computing Machinery, 884–895.

[5] Gebrekiros Gebreyesus Gebremariam, J Panda, and S Indu. 2023. Blockchain-based secure localization against malicious nodes in IoT-based wireless sensor networks using federated learning. *Wireless communications and mobile computing* 2023, 1 (2023), 8068038.

[6] Jhony H Giraldo, Arif Mahmood, Belmar Garcia-Garcia, Dorina Thanou, and Thierry Bouwmans. 2022. Reconstruction of time-varying graph signals via Sobolev smoothness. *IEEE Transactions on Signal and Information Processing over Networks* 8 (2022), 201–214.

[7] Abhishek Grover and Brejesh Lall. 2021. A recursive method for estimating missing data in spatio-temporal applications. *IEEE Transactions on Industrial Informatics* 18, 4 (2021), 2714–2723.

[8] Jing Jiang, Chunxu Zhang, Honglei Zhang, Zhiwei Li, Yidong Li, and Bo Yang. 2025. A Tutorial of Personalized Federated Recommender Systems: Recent Advances and Future Directions. Association for Computing Machinery, New York, NY, USA, 21–24. doi:10.1145/3701716.3715860

[9] Aditya Pribadi Kalapaaking, Ibrahim Khalil, and Mohammed Atiquzzaman. 2023. Smart policy control for securing federated learning management system. *IEEE Transactions on Network and Service Management* 20, 2 (2023), 1600–1611.

[10] Bin Li, Yulong Zou, Jia Zhu, and Weifeng Cao. 2021. Impact of hardware impairment and co-channel interference on security-reliability trade-off for wireless sensor networks. *IEEE Transactions on Wireless Communications* 20, 11 (2021), 7011–7025.

[11] Qinbin Li, Zeyi Wen, Zhaomin Wu, Sixu Hu, Naibo Wang, Yuan Li, Xu Liu, and Bingsheng He. 2021. A survey on federated learning systems: Vision, hype and reality for data privacy and protection. *IEEE Transactions on Knowledge and Data Engineering* 35 (2021), 3347–3366.

[12] Yichen Li, Yijing Shan, Yi Liu, Haozhao Wang, Wei Wang, Yi Wang, and Ruixuan Li. 2025. Personalized Federated Recommendation for Cold-Start Users via Adaptive Knowledge Fusion. Association for Computing Machinery, New York, NY, USA, 2700–2709. doi:10.1145/3696410.3714635

[13] Feng Liang, Weike Pan, and Zhong Ming. 2021. Fedrec++: Lossless federated recommendation with explicit feedback. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 35. 4224–4231.

[14] Guanyu Lin, Feng Liang, Weike Pan, and Zhong Ming. 2020. Fedrec: Federated recommendation with explicit feedback. *IEEE Intelligent Systems* 36 (2020), 21–30.

[15] Yujie Lin, Pengjie Ren, Zhumin Chen, Zhaochun Ren, Dongxiao Yu, Jun Ma, Maarten de Rijke, and Xiuzhen Cheng. 2020. Meta matrix factorization for federated rating predictions. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*. 981–990.

[16] Jingping Liu, Xiujuan Du, and Long Jin. 2024. A Localization Algorithm for Underwater Acoustic Sensor Networks With Improved Newton Iteration and Simplified Kalman Filter. *IEEE Transactions on Mobile Computing* 23, 12 (2024), 14459–14470. doi:10.1109/TMC.2024.3443992

[17] Jingxiang Liu, Jian Peng, Wenzheng Xu, Weifa Liang, Tang Liu, Xi Peng, Zichuan Xu, Zheng Li, and Xiaohua Jia. 2023. Maximizing Sensor Lifetime via Multi-node Partial-Charging on Sensors. *IEEE Transactions on Mobile Computing* 22, 11 (2023), 6571–6584. doi:10.1109/TMC.2022.3200070

[18] Langming Liu, Wanyu Wang, Xiangyu Zhao, Zijian Zhang, Chunxu Zhang, Shanru Lin, Yiqi Wang, Lixin Zou, Zitao Liu, Xuetao Wei, Hongzhi Yin, and Qing Li. 2024. Efficient and Robust Regularized Federated Recommendation. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (CIKM '24)*. Association for Computing Machinery, New York, NY, USA, 1452–1461. doi:10.1145/3627673.3679682

[19] Quande Liu. 2022. *Multi-center Medical Image Computing with Robust Federated Learning and Beyond*. Ph. D. Dissertation. Advisor(s) Wing, Fu, Chi and Ann, Heng, Pheng. AAI30159745.

[20] Yueliang Liu, Wenbin Guo, Kangyong You, Lei Zhao, Tao Peng, and Wenbo Wang. 2020. Graph Learning for Spatiotemporal Signals With Long- and Short-Term Characterization. *IEEE Transactions on Signal and Information Processing over Networks* 6 (2020), 699–713. doi:10.1109/TSIPN.2020.3038475

[21] Zhiwei Liu, Liangwei Yang, Ziwei Fan, Hao Peng, and Philip S Yu. 2022. Federated social recommendation with graph neural network. *ACM Transactions on Intelligent Systems and Technology (TIST)* 13 (2022), 1–24.

[22] Xin Luo, Ye Yuan, Sili Chen, Nianyin Zeng, and Zidong Wang. 2022. Position-Transitional Particle Swarm Optimization-Incorporated Latent Factor Analysis. *IEEE Transactions on Knowledge and Data Engineering* 34 (2022), 3958–3970. doi:10.1109/TKDE.2020.3033324

[23] Zhihan Lv, Liang Qiao, Amit Kumar Singh, and Qingjun Wang. 2021. AI-empowered IoT security for smart cities. *ACM Transactions on Internet Technology* 21, 4 (2021), 1–21.

[24] Xianghui Mao, Kai Qiu, Tiejian Li, and Yuantao Gu. 2018. Spatio-temporal signal recovery based on low rank and differential smoothness. *IEEE Transactions on Signal Processing* 66, 23 (2018), 6281–6296.

[25] H. Brendan McMahan, Eider Moore, Daniel Ramage, and Blaise Agüera y Arcas. 2016. Federated Learning of Deep Networks using Model Averaging. *CoRR* abs/1602.05629 (2016). arXiv:1602.05629 http://arxiv.org/abs/1602.05629

[26] Jifei Miao and Kit Ian Kou. 2021. Color image recovery using low-rank quaternion matrix completion algorithm. *IEEE Transactions on Image Processing* 31 (2021), 190–201.

[27] Antonio Ortega, Pascal Frossard, Jelena Kovačević, José M. F. Moura, and Pierre Vandergheynst. 2018. Graph Signal Processing: Overview, Challenges, and Applications. *Proc. IEEE* 106, 5 (2018), 808–828. doi:10.1109/JPROC.2018.2820126

[28] Yayu Peng, Wei Qiao, and Liyan Qu. 2021. Compressive sensing-based missing-data-tolerant fault detection for remote condition monitoring of wind turbines. *IEEE Transactions on Industrial Electronics* 69, 2 (2021), 1937–1947.

[29] Xinglin Piao, Yongli Hu, Yanfeng Sun, Baocai Yin, and Junbin Gao. 2014. Correlated Spatio-Temporal Data Collection in Wireless Sensor Networks Based on Low Rank Matrix Approximation and Optimized Node Sampling. *Sensors* 14, 12 (2014), 23137–23158. doi:10.3390/s141223137

[30] Seno Adi Putra, Bambang Riyanto Trilaksono, Muhammad Riyansyah, and Dina Shona Laila. 2020. Multiagent architecture for bridge capacity measurement system using wireless sensor network and weight in motion. *IEEE Transactions on Instrumentation and Measurement* 70 (2020), 1–14.

[31] Adnan Qayyum, Kashif Ahmad, Muhammad Ahtazaz Ahsan, Ala Al-Fuqaha, and Junaid Qadir. 2022. Collaborative federated learning for healthcare: Multi-modal covid-19 diagnosis at the edge. *IEEE Open Journal of the Computer Society* 3 (2022), 172–184.

[32] Kai Qiu, Xianghui Mao, Xinyue Shen, Xiaohan Wang, Tiejian Li, and Yuantao Gu. 2017. Time-varying graph signal reconstruction. *IEEE Journal of Selected Topics in Signal Processing* 11, 6 (2017), 870–883.

[33] Sourish Sarkar and Rajiv Ranjan Sahay. 2021. A non-local superpatch-based algorithm exploiting low rank prior for restoration of hyperspectral images. *IEEE Transactions on Image Processing* 30 (2021), 6335–6348.

[34] Jingwei Song, Shaobo Xia, Jun Wang, Mitesh Patel, and Dong Chen. 2021. Uncertainty quantification of hyperspectral image denoising frameworks based on sliding-window low-rank matrix approximation. *IEEE Transactions on Geoscience and Remote Sensing* 60 (2021), 1–12.

[35] Jingwei Song, Shaobo Xia, Jun Wang, Mitesh Patel, and Dong Chen. 2022. Uncertainty Quantification of Hyperspectral Image Denoising Frameworks Based on Sliding-Window Low-Rank Matrix Approximation. *IEEE Transactions on Geoscience and Remote Sensing* 60 (2022), 1–12. doi:10.1109/TGRS.2021.3065570

[36] Tinghui Wang and Diane J Cook. 2020. sMRT: Multi-resident tracking in smart homes with sensor vectorization. *IEEE transactions on pattern analysis and machine intelligence* 43, 8 (2020), 2809–2821.

[37] Chuhan Wu, Fangzhao Wu, Yang Cao, Yongfeng Huang, and Xing Xie. 2021. Fedgnn: Federated graph neural network for privacy-preserving recommendation. (2021).

[38] Di Wu, Zechao Li, Zhikai Yu, Yi He, and Xin Luo. 2025. Robust Low-Rank Latent Feature Analysis for Spatiotemporal Signal Recovery. *IEEE Transactions on Neural Networks and Learning Systems* 36 (2025), 2829–2842. doi:10.1109/TNNLS.2023.3339786

[39] Di Wu, Mingsheng Shang, Xin Luo, and Zidong Wang. 2022. An L1-and-L2-Norm-Oriented Latent Factor Model for Recommender Systems. *IEEE Transactions on Neural Networks and Learning Systems* 33 (2022), 5775–5788. doi:10.1109/TNNLS.2021.3071392

[40] Ting Xie, Shutao Li, and Jibao Lai. 2021. Adaptive rank and structured sparsity corrections for hyperspectral image restoration. *IEEE Transactions on Cybernetics* 52, 9 (2021), 8729–8740.

[41] Peipei Xu, Wenjie Ruan, Quan Z. Sheng, Tao Gu, and Lina Yao. 2017. Interpolating the Missing Values for Multi-Dimensional Spatial-Temporal Sensor Data: A Tensor SVD Approach. In *Proceedings of the 14th EAI International Conference on Mobile and Ubiquitous Systems: Computing, Networking and Services*. Association for Computing Machinery, New York, NY, USA, 442–451. doi:10.1145/3144457.3144474

[42] Liu Yang, Yonina C. Eldar, Haifeng Wang, Kai Kang, and Hua Qian. 2021. An ADMM-Net for Data Recovery in Wireless Sensor Networks. In *2020 28th European Signal Processing Conference (EUSIPCO)*. 1712–1716.

[43] Qiang Yang, Yang Liu, Tianjian Chen, and Yongxin Tong. 2019. Federated machine learning: Concept and applications. *ACM Transactions on Intelligent Systems and Technology (TIST)* 10 (2019), 1–19.

[44] Xianyao You, Ximeng Liu, Nan Jiang, Jianping Cai, and Zuobin Ying. 2023. Reschedule Gradients: Temporal Non-IID Resilient Federated Learning. *IEEE*

Internet of Things Journal 10, 1 (2023), 747–762. doi:10.1109/JIOT.2022.3203233
[45] Linan Yue, Qi Liu, Yawen Li, Fangzhou Yao, Weibo Gao, and Junping Du. 2025. Empowering Federated Graph Rationale Learning with Latent Environments. Association for Computing Machinery, New York, NY, USA, 3863–3873. doi:10.1145/3696410.3714929
[46] Shanhao Zhan, Lianfen Huang, Gaoyu Luo, Shaolong Zheng, Zhibin Gao, and Han-Chieh Chao. 2025. A Review on Federated Learning Architectures for Privacy-Preserving AI: Lightweight and Secure Cloud–Edge–End Collaboration. *Electronics* 14, 13 (2025), 2512.
[47] Chen Zhang, Yu Xie, Hang Bai, Bin Yu, Weihong Li, and Yuan Gao. 2021. A survey on federated learning. *Knowledge-Based Systems* 216 (2021), 106775.
[48] Xiaoyue Zhang, Jingfei He, Yunpei Li, Yue Chi, and Yatong Zhou. 2021. Recovery of Corrupted Data in Wireless Sensor Networks Using Tensor Robust Principal Component Analysis. *IEEE Communications Letters* 25, 10 (2021), 3389–3393. doi:10.1109/LCOMM.2021.3097158
[49] Ziqi Zhao, Dong F Wang, Yipeng Hou, Toshihiro Itoh, and Ryutaro Maeda. 2020. Noninvasive passive measurement of multiphase currents in IoT. *IEEE Transactions on Industrial Electronics* 68, 12 (2020), 12860–12870.
[50] Ligeng Zhu, Zhijian Liu, and Song Han. 2019. Deep leakage from gradients. *Advances in neural information processing systems* 32 (2019).
[51] Tongxin Zhu, Jianzhong Li, Hong Gao, Yingshu Li, and Zhipeng Cai. 2023. AoI Minimization Data Collection Scheduling for Battery-Free Wireless Sensor Networks. *IEEE Transactions on Mobile Computing* 22, 3 (2023), 1343–1355. doi:10.1109/TMC.2021.3106013

A Appendix

This section first introduces the algorithmic design and analyzes its computational complexity. We then describe the experimental setup and provide detailed accounts of the baseline methods. Finally, we present additional results on datasets not included in the main text.

A.1 Algorithm Design and Complexity Analysis

The complete pseudocode of FLFL is given in Algorithm 1. For each sensor i, the worst time complexity is $\Theta(T \times N \times k)$. For the server, the worst time complexity is $\Theta(T \times M \times k)$. For the communication complexity, the main communication is that each sensor i sends ∇q to the server and receives the model parameter Q from the server. The gradient contains $8 \times k \times N$ bytes of information and the model parameter contains $8 \times k \times N$ bytes of information. Hence, for each sensor i at each iteration, the communication complexity is $16 \times k \times N$ bytes. For the server at each iteration, the communication complexity is $8 \times k \times N$ bytes.

Algorithm 1: Alternative Stochastic Gradient Descent in FLFL

Region partitioning: obtain H regions according to Eq. (5);
Initialization: Server randomly initializes the global latent matrix Q;
Each sensor initializes its latent vector;
Server-side:;
while $t \leq T$ ***and*** *not converged* **do**
  for $l = 1$ **to** M **do**
    Randomly sample a sensor i from $\mathcal{S}$;
    Call SensorUpdate$(Q, i, \eta, z_1, z_2, \lambda, t)$;
    for $j = 1$ **to** N **do**
      for $r = 1$ **to** k **do**
        Update Q according to Eq. (13);

Sensor-side (procedure SensorUpdate): Initialization: Download model parameters from the server;
Download latent feature vectors of neighboring sensors within the same region;
foreach $y_{i,j} \in Y_i^K$ **do**
  for $k' = 1$ **to** k **do**
    Compute gradients $\nabla(p_{i,\cdot}^h)^t$ and $\nabla q_{i,\cdot}^t$ according to Eq. (10) and Eq. (11);
    Update $(p_{i,\cdot}^h)^t$ according to Eq. (12);
Upload gradient ∇q^t to the server;

Table 3: Baseline models for comparison.

Model	Description
RFRec[18]	This federated framework efficiently optimizes the empirical risk minimization problem. (CIKM,2024)
FeSoG[21]	A federated learning framework for social recommendation using GNN, relational attention, and pseudo-labeling for privacy and personalization. (ACM TIST, 2022)
FedMF[3]	Federated matrix factorization with homomorphic encryption. (IEEE IS 2020)
FedRec++[13]	Lossless federated recommendation that assigns denoising clients to remove noise in a privacy-aware manner. (AAAI 2021)
MetaMF[15]	Neural architecture for federated collaborative ranking via memory-based attention. (SIGIR 2020)

A.2 Experimental Details.

To simulate missing data in WSNs, a portion of the complete dataset is randomly selected to form the training set, and the remaining data serve as the testing set. Within the training set, half of the data are used for model training and the other half for validation and hyperparameter tuning. After determining the optimal hyperparameters, the model is retrained using all data in the training split. The maximum number of iterations is set to 3000, and training stops early if the error difference between two consecutive iterations satisfies $\left|\mathrm{err}^{(t)} - \mathrm{err}^{(t-1)}\right| < 10^{-6}$. Each experiment is repeated five times, and the average results are reported. All experiments are conducted on a PC with an Intel Core i7 CPU (2.1 GHz) and 32 GB of RAM.

A.3 Baseline Description

A detailed description of the baseline method is provided below; see Table 3.

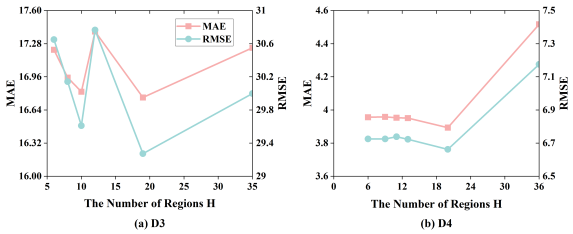

Figure 7: The MAE/RMSE of FLFL with different H on D3 and D4

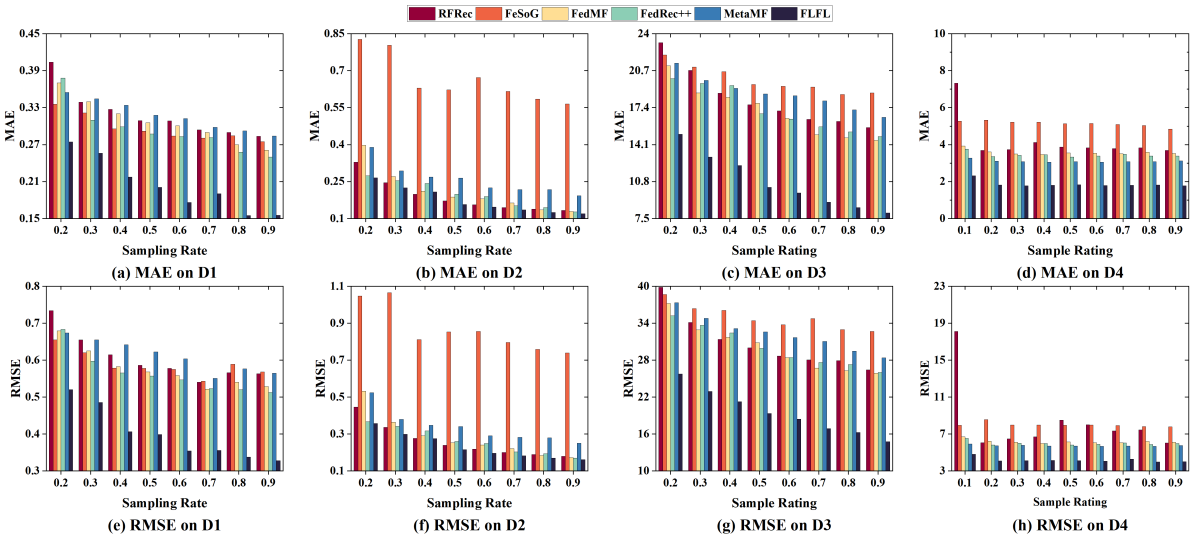

Figure 6: The comparison results of recovery accuracy with different sampling rates from 0.1 to 0.9 on all datasets.

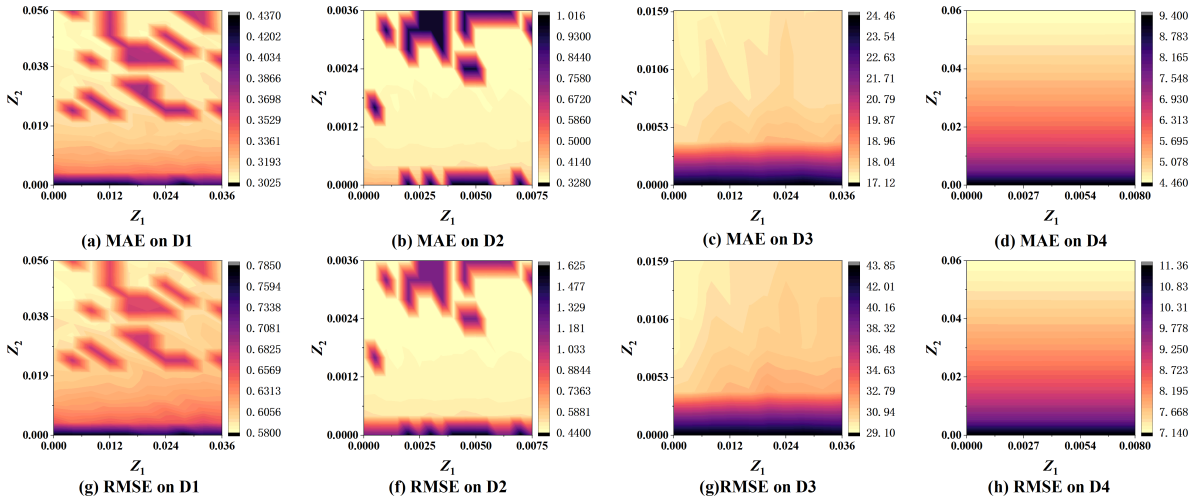

Figure 8: The MAE/RMSE of FLFL with different z_1 and z_2 on all Datasets

A.4 Additional Experiment Results

A.4.1 Complete Comparison Performance. Figure 6 presents the complete comparison results. As the sampling rate increases, the MAE and RMSE of FLFL decrease steadily. Across all datasets and sampling-rate settings, FLFL consistently achieves the lowest MAE/RMSE among federated models. These consistent results indicate that FLFL offers superior reconstruction accuracy compared with other federated methods.

A.4.2 Sensitivity to the Number of Partitioned Regions H. This subsection reports additional results on datasets not shown in the main text. We analyze the sensitivity of the number of regions H. As shown in Figure 7, when H increases, MAE and RMSE decrease at first. They reach an optimum within a certain range. Then they rise. This pattern suggests that a careful choice of H is essential for better reconstruction accuracy.

A.4.3 Sensitivity to spatio-temporal regularization. The sampling rate is fixed at 0.1. Holding other hyper-parameters constant, we perform a grid search over z_1 and z_2 to evaluate their impact on the recovery accuracy of FLFL. Figure 8 reports FLFL's accuracy on all datasets under different z_1 and z_2 values. The results show that z_1 and z_2 substantially affect performance: as either z_1 or z_2 increase, MAE/RMSE decreases until reaching an optimum, after which both metrics begin to rise. For example, on D1, once $z_1 = 0.0015$ and $z_2 = 0.02$, further increases in either hyperparameter lead to higher MAE and RMSE. We also note that z_1 exerts only a minor influence on D1–D3, likely because relatively sparse sensor layouts attenuate spatial correlations in these datasets. Taken together, these

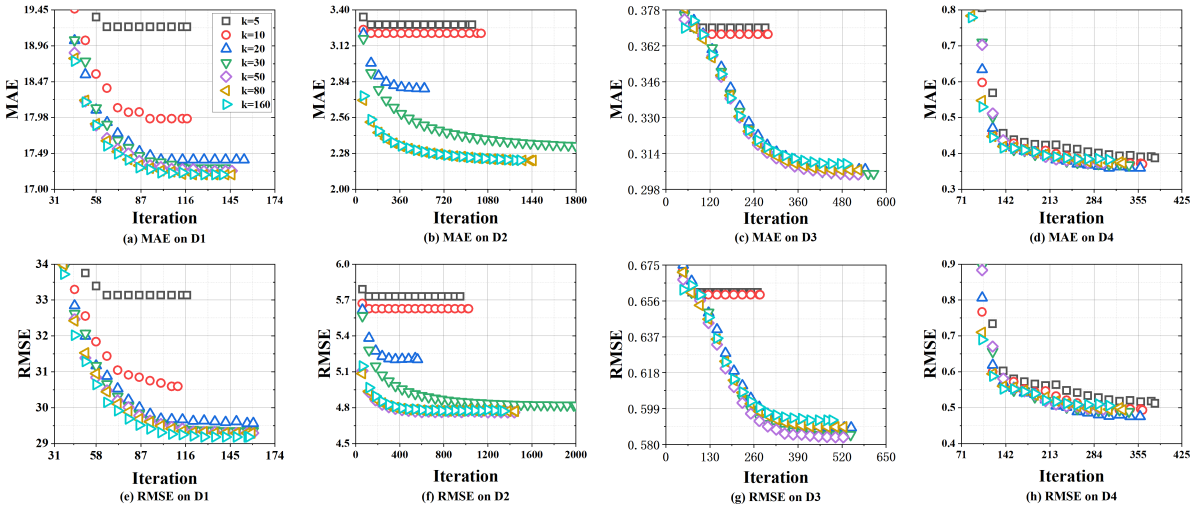

Figure 9: The MAE/RMSE of FLFL with different k from 5 to 160 results on all Datasets

results highlight the necessity of properly tuning spatio-temporal regularization to maximize the recovery accuracy of FLFL.

A.4.4 The impact of latent feature dimension k. We study the effect of k at a sampling rate of 0.1. Figure 9 shows results on D1 and D2 with $k \in \{5, \ldots, 160\}$; Overall, MAE/RMSE decreases as k grows, confirming that a larger k typically improves FLFL's recovery accuracy. However, beyond a moderate threshold (e.g., $k \gtrsim 20$), the gains taper off and may even reverse, likely due to overfitting under low sampling (e.g., 0.1). Moreover, the computational cost of FLFL scales approximately linearly with k. Balancing accuracy and efficiency, we recommend $k = 30$ in practice.

A.4.5 FLFL vs. non-federated FLFL performance. With the sampling rate fixed at 0.1, we compare federated and non-federated models using two normalized indices: the mean difference (MD) and the standard-deviation range (STDR)[14]. Results are reported in Table 4. Overall, the gap between federated and non-federated FLFL is negligible. On the D3 dataset, the federated FLFL even achieves lower MAE and RMSE. These findings indicate that federation preserves accuracy while providing privacy benefits, with no meaningful performance penalty.

Table 4: Comparison between FLFL and non-federated FLFL.

Dataset	Metric	TRSS	FLFL		MD	STDR
			non-fed	fed		
D1	MAE	$0.347_{\pm 0.036}$	$0.305_{\pm 0.002}$	$0.305_{\pm 0.002}$	0.00%	1.31%
	RMSE	$0.648_{\pm 0.056}$	$0.434_{\pm 0.002}$	$0.434_{\pm 0.002}$	0.00%	0.92%
D2	MAE	$0.926_{\pm 0.071}$	$0.323_{\pm 0.001}$	$0.324_{\pm 0.002}$	0.30%	0.92%
	RMSE	$1.265_{\pm 0.034}$	$0.432_{\pm 0.001}$	$0.433_{\pm 0.011}$	0.23%	2.77%
D3	MAE	$18.479_{\pm 1.256}$	$18.057_{\pm 0.312}$	$18.023_{\pm 0.290}$	-0.18%	3.29%
	RMSE	$31.086_{\pm 1.341}$	$30.514_{\pm 0.006}$	$30.479_{\pm 0.128}$	-0.11%	0.43%
D4	MAE	$3.024_{\pm 0.664}$	$2.308_{\pm 0.005}$	$2.307_{\pm 0.029}$	-0.04%	1.47%
	RMSE	$5.735_{\pm 0.659}$	$4.806_{\pm 0.014}$	$4.806_{\pm 0.024}$	0.00%	0.79%

A.4.6 Differential Privacy in Partitioned Regions. To protect the data privacy of sensors within each partitioned region and to prevent privacy leakage caused by sharing latent feature vectors, we incorporate the principle of (ϵ,δ)-differential privacy[1]. Specifically, each sensor injects Gaussian noise into its latent feature vector before sharing it, ensuring that sensors within the same region cannot recover the true values of others. In this way, even though latent features are exchanged within a region, individual sensor data remain protected, effectively preserving data privacy.

In the experimental setup, we fix the clipping threshold at $c = 0.1$ and set the noise multiplier nm to $\{0, 1, 5, 10\}$. Under the sampling rate of 0.1, we systematically compare the recovery capability of FLFL across different nm settings, with the results summarized in Table 5. The experimental results show that as the noise multiplier nm increases, the representation capability of FLFL gradually declines, while its privacy protection capability is correspondingly enhanced. Nevertheless, even under strong privacy constraints, FLFL maintains a competitive level of recovery performance.

Table 5: Performance of Differential Privacy in Partitioned Region

Dataset	Metric	$nm = 0$	$nm = 1$	$nm = 5$	$nm = 10$
D1	MAE	$\mathbf{0.305}_{\pm 0.002}$	$0.305_{\pm 0.033}$	$0.306_{\pm 0.029}$	$0.308_{\pm 0.038}$
	RMSE	$\mathbf{0.434}_{\pm 0.002}$	$0.434_{\pm 0.066}$	$0.436_{\pm 0.051}$	$0.440_{\pm 0.054}$
D2	MAE	$\mathbf{0.324}_{\pm 0.002}$	$0.347_{\pm 0.001}$	$0.378_{\pm 0.001}$	$0.433_{\pm 0.002}$
	RMSE	$\mathbf{0.433}_{\pm 0.011}$	$0.465_{\pm 0.000}$	$0.508_{\pm 0.000}$	$0.580_{\pm 0.002}$
D3	MAE	$\mathbf{18.023}_{\pm 0.290}$	$18.093_{\pm 0.229}$	$18.141_{\pm 0.361}$	$18.327_{\pm 0.300}$
	RMSE	$\mathbf{30.479}_{\pm 0.128}$	$30.879_{\pm 0.171}$	$31.079_{\pm 0.150}$	$31.186_{\pm 0.142}$
D4	MAE	$\mathbf{2.307}_{\pm 0.029}$	$2.508_{\pm 0.024}$	$2.612_{\pm 0.017}$	$2.612_{\pm 0.027}$
	RMSE	$\mathbf{4.806}_{\pm 0.024}$	$4.949_{\pm 0.033}$	$4.992_{\pm 0.016}$	$4.993_{\pm 0.027}$